

Smart Agricultural Technology

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An AI-Powered Agricultural Decision Support System for Smallholder Farmers in India — Multi-Model Weather Forecasting, Real-Time Hazard Alerting, and Mobile-Ready Crop Advisory

Abstract — Weather variability and climate extremes increasingly threaten India's 600 million agriculture-dependent population, yet no end-to-end, deployable decision-support tool exists for the country's 86% smallholder farmer demographic. This paper presents AgriTech Platform, a full-stack AI-powered agricultural advisory platform that operationalises weather machine learning (ML) into actionable, farmer-centric guidance. Using a purpose-built ERA5 reanalysis dataset spanning 39 Indian cities across 14 years (2010–2023, 199,407 daily records), three complementary models are developed and evaluated across six metrics: Seasonal Autoregressive Integrated Moving Average (SARIMA) as a statistical baseline, and Random Forest (RF) Regressor and Classifier as primary predictors. The RF Regressor outperforms SARIMA on all metrics: RMSE 1.003 vs. 1.664°C, MAE 0.746 vs. 1.335°C, R^2 0.961 vs. 0.511, MAPE 2.47% vs. 4.43%, and Directional Accuracy 59.9% vs. 40.4%. Note that SARIMA is evaluated on weekly-aggregated single-city (Mumbai) data while the RF Regressor is evaluated on daily multi-city data; the comparison is therefore indicative of relative model characteristics rather than a controlled head-to-head benchmark. The RF model maintains RMSE below 1.0°C for seven-day forecast windows, directly supporting crop-cycle planning horizons. The RF Classifier achieves 84.6% accuracy and 78% Rain-class recall for next-day rainfall prediction. Beyond model performance, the central contribution is the integration of these models into a working Django-based web platform with a fully functional proof-of-concept mock version available on GitHub at [https://github.com/\[anonymous\]/WeatherPredictionSystem](https://github.com/[anonymous]/WeatherPredictionSystem). The platform delivers 15-day RF weather forecasts, compound disease event detection, MSP-based crop revenue projections, and an AI farm assistant chatbot. The long-term product vision is a multilingual, voice-assisted mobile application for low-literacy farmers on 2G networks establishing a platform as a scalable AgriTech IP product at the intersection of AI, climate data, and food security.

Keywords — Machine Learning | weather forecasting | agricultural decision support | ERA5 | smallholder farmers | crop advisory | India

I. INTRODUCTION

India's agricultural sector is the livelihood of nearly 600 million people, yet it remains one of the world's most exposed to climate risk. A significant majority of these farmers approximately 86% cultivate plots of under 2 hectares [11], making their yields highly sensitive to temperature swings, irregular monsoons, and weather extremes that rain-fed agriculture simply cannot absorb. Despite decades of effort and investment in numerical weather prediction, a fundamental gap persists: forecasts continue to be produced at scales and in formats that leave smallholder farmers without the specific, timely guidance they need for day-to-day decisions around irrigation, sowing, and crop protection.

Machine learning has opened meaningful new avenues in weather forecasting. A range of approaches like Long Short-Term Memory (LSTM) networks, Convolutional Neural Networks (CNN), Support Vector Regression (SVR), and ensemble methods like

Random Forest (RF) have been applied to meteorological data, each offering ways to capture patterns that linear statistical models routinely miss [1][3]. Yet the bulk of this research stops at the model level. Most studies are built around performance benchmarking and never seriously engage with how predictions might reach a farmer who could actually benefit from them. There is also a persistent evaluation blind spot: the field has largely converged on reporting only MSE and RMSE, leaving out complementary measures such as MAE, MAPE, R^2 , and Directional Accuracy that together offer a much more complete picture of real-world forecasting utility.

This paper addresses these limitations through a working proof-of-concept AgriTech platform that integrates SARIMA and Random Forest models into a full-stack agricultural advisory system. The mock version, with all features implemented, is publicly available on GitHub. The main contributions are:

- Multi-model forecasting with full six-metric evaluation: comparative evaluation of SARIMA

and RF across MSE, RMSE, MAE, MAPE, R^2 , and Directional Accuracy on 199,407 ERA5 records across 39 Indian cities (2010–2023).

- Working platform as proof of concept: a fully functional mock version of AgriTech Platform deployed on GitHub demonstrating the complete system pipeline.
- Actionable agricultural intelligence: 15-day RF forecasts, compound disease event detection (novel multi-parameter alert), MSP-based crop revenue projections, and an AI chatbot.
- Multi-regional open dataset: release of IndiaWeatherDataset_2010_2023 with 199,407 ERA5 records across 39 cities and 15 Indian states under MIT licence on GitHub.
- Product roadmap: modular architecture extensible to IMD sensor integration and a multilingual, voice-assisted mobile application targeting 2G-capable, low-literacy farmers.

II. LITERATURE SURVEY

Research at the intersection of weather forecasting and agricultural support has developed along three main lines: deep learning applied to time series prediction, ensemble and classical machine learning methods, and systems specifically designed to support agricultural decision-making.

Deep learning models, particularly hybrid CNN-LSTM architectures, have demonstrated strong capability in modelling complex temporal

dependencies in meteorological data. Recent work on Delhi temperature prediction using CNN-LSTM achieved an RMSE of 1.80°C [1]. While these approaches offer improved predictive performance, they are computationally intensive and lack practical deployment pathways which are a critical barrier for resource-constrained AgriTech applications.

Random Forest-based models serve as robust and interpretable alternatives, achieving MSE as low as 0.750 and R^2 up to 0.97 for non-linear meteorological prediction [7]. Comparative studies involving RF, SVR, and Ridge Regression confirm RF's superiority in capturing non-linear relationships [2]. Agriculture-focused ML research highlights the importance of weather variables in crop yield estimation and disease prediction [4][9].

Three shortcomings run consistently through the existing literature. Most papers limit their evaluation to MSE and RMSE, leaving a large share of performance information unreported. Very few studies combine multiple modelling approaches within a single unified framework. Most importantly, no published work has produced a functioning, end-to-end system that takes weather machine learning outputs and turns them into actionable agricultural guidance for Indian smallholder farmers. Table I provides a structured comparison of the relevant landscape.

TABLE I. COMPARATIVE REVIEW OF EXISTING APPROACHES — WEATHER ML AND AGRICULTURAL DECISION SUPPORT

Ref.	Algorithm	Key Contribution	Deployed?	Dataset/Region	Gap vs This Work	Metrics
[1]	CNN-LSTM	Hybrid deep learning, temp prediction	No	Delhi, India	No deployment, no agri advisory	RMSE 1.80
[2]	RF, SVR, Ridge	Multi-model ML comparison	No	Tennessee, USA	No agricultural application	RMSE 3.0
[3]	Naïve Bayes, χ^2	Statistical learning + feature selection	No	Unspecified	Data-quality sensitive, no system	Acc 93%
[4]	Review	Crop growth pattern survey	N/A	N/A	No new model or deployment	N/A
[5]	ANN, RF	ML for missing precipitation imputation	No	South Korea	Single variable, no advisory output	RF Acc 84%
[6]	Linear/Func. Reg.	Short vs. long-term baseline forecasting	No	Unspecified	Simple models, no ensemble	N/A
[7]	Random Forest	Ensemble methods, non-linear weather	No	Unspecified	Single variable, no system	MSE 0.750, R^2 0.97
[8]	RNN, Multi-Reg.	Simultaneous multi-variable prediction	No	Unspecified	No agricultural output	N/A

[9]	Linear Reg., DT	ML for agricultural planning	No	Unspecified	Monthly data, no deployment	MSE 0.739
[10]	MLR, SVM, Lasso	Long-term rainfall prediction	No	India 1901–2015	Single variable, no system	SVR R ² 0.996
Proposed	RF + SARIMA	Multi-city ERA5, 6-metric eval, deployed	Yes (GitHub)	39 cities, India, 2010–2023	—	RMSE 1.003, R² 0.961

III. DATASET AND PREPROCESSING

A. Dataset Construction

The study draws on two datasets. The primary one, IndiaWeatherDataset_2010_2023, was assembled by querying the ERA5 reanalysis archive through the Open-Meteo historical API [20], targeting 48 Indian cities selected to cover all major agro-climatic zones. API rate limits were handled using exponential backoff, and 39 of the 48 cities were successfully retrieved, producing 199,407 daily records spanning 1 January 2010 through 31 December 2023. Every record includes six meteorological variables: maximum temperature (temp_max, °C), minimum temperature (temp_min, °C), mean temperature (temp_mean, °C), relative humidity (humidity, %), maximum wind speed (wind_max, km/h), and total daily precipitation (precip, mm). No missing values were found in any column.

Spread across 15 states, the 39 cities capture the full breadth of India's agro-climatic variation from the arid summer heat of Rajasthan (Jodhpur records a mean April maximum of 39.2°C) to the coastal humidity of Kerala (Kochi at 31.5°C) and the heavy monsoon conditions of Assam (Guwahati). ERA5 reanalysis was preferred over ground-based station data primarily because of its spatial completeness and unbroken temporal coverage; ground observation networks in India are unevenly distributed and prone to data gaps that would undermine a 14-year training set. A secondary dataset HistoricalData.csv, containing 4,227 records from Mumbai spanning 2010 to 2021 provided the soil profiles, disease labels, and extended weather variables needed to calibrate the thresholds driving the alert engine.

B. Exploratory Data Analysis

The seasonal cycle analysis (Fig. 2) reveals a clear bimodal temperature pattern across the 39 cities, with mean maximum temperatures peaking through May and June before falling during the June to September monsoon period. Rainfall is heavily concentrated in

the Kharif season, accounting for roughly 80% of the annual total across central and north India. When city-level climate is plotted by temperature and humidity (Fig. 3), three agro-climatic clusters emerge: a hot-arid group (Jodhpur, Jaipur, Ahmedabad), a hot-humid coastal group (Kochi, Visakhapatnam) where April humidity sits above the 82% disease-risk threshold, and a moderate inland group in between. Long-term trend analysis (Fig. 4) detects a statistically significant warming rate of +0.021°C per year across the 39-city aggregate. Seasonal decomposition of Mumbai's weekly maximum temperature (Fig. 5) confirms a consistent 12-month seasonal cycle alongside a cumulative warming of approximately +0.3°C over the 14-year study window.

C. Feature Engineering

The raw ERA5 variables were transformed into a richer set of temporal features to capture the patterns most relevant to next-day temperature forecasting. Temperature lags were computed at three intervals which are 1 day (temp_lag1), 7 days (temp_lag7), and 14 days (temp_lag14), alongside lagged humidity (hum_lag1) and wind speed (wind_lag1). Rolling means and standard deviations over 7-day and 30-day windows were added to represent short-term atmospheric persistence and variability. Rather than treating month as an ordinal variable, sine and cosine transformations ($\sin(2\pi m/12)$, $\cos(2\pi m/12)$) were applied to preserve the inherently circular structure of the calendar year. Each record in the final feature matrix contains 18 variables.

D. Train / Validation / Test Split

The dataset was split chronologically to preserve temporal structure and prevent data leakage. Records from 2010 to 2019 formed the training set, 2020–2021 the validation set, and 2022–2023 the test set (approximately 70/15/15 split across 14 years). The binary rainfall label for the RF Classifier was defined as precipitation > 2.5 mm on the following day, yielding a moderately imbalanced dataset (approximately 23% positive class). Class imbalance was addressed via class_weight='balanced' in the RF Classifier, prioritising Rain-class recall.

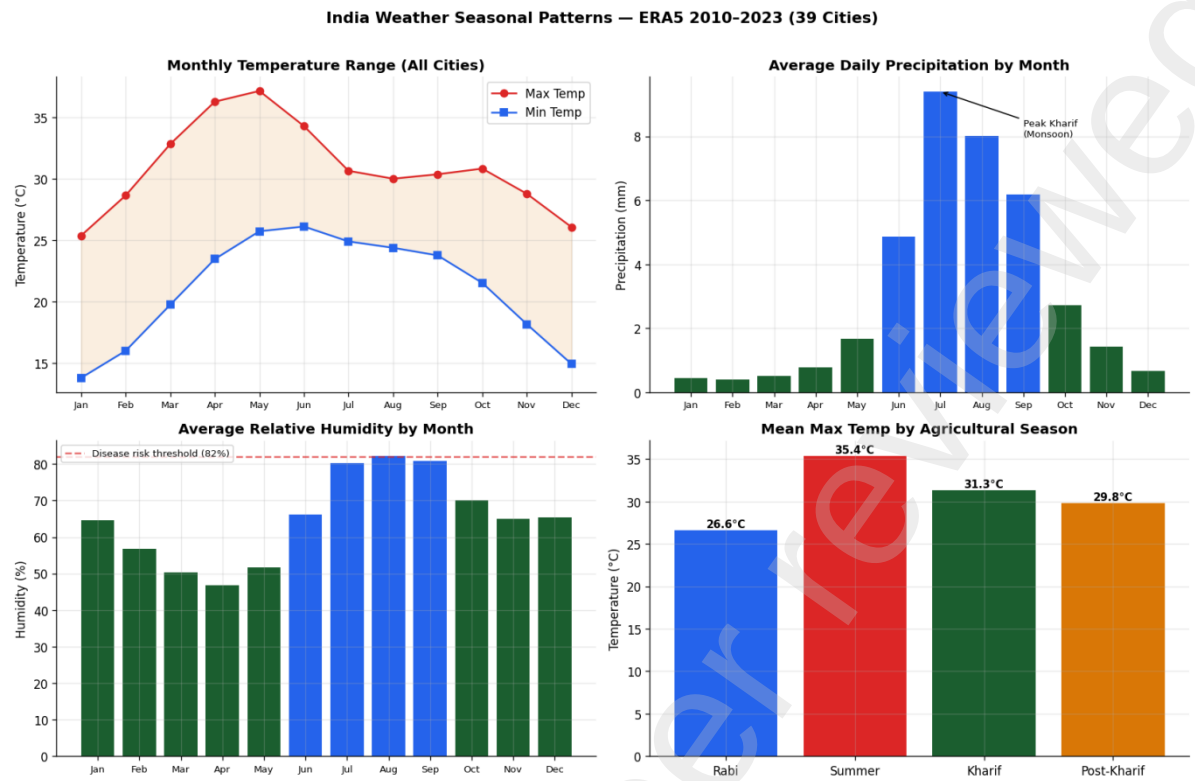


FIGURE 2. ERA5 Dataset: National Seasonal Cycle - Temperature, Precipitation, and Humidity (39 Cities, 2010–2023)

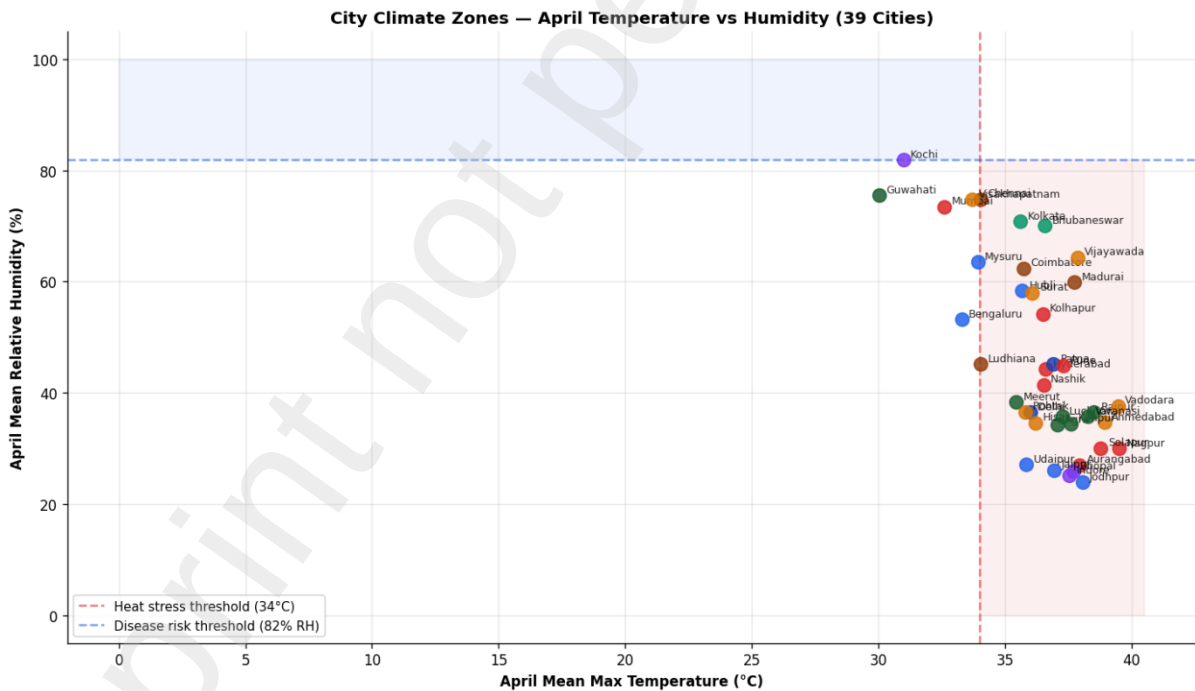


FIGURE 3. City Agro-Climature Scatter: April Mean Max Temperature vs. Relative Humidity (39 Cities)

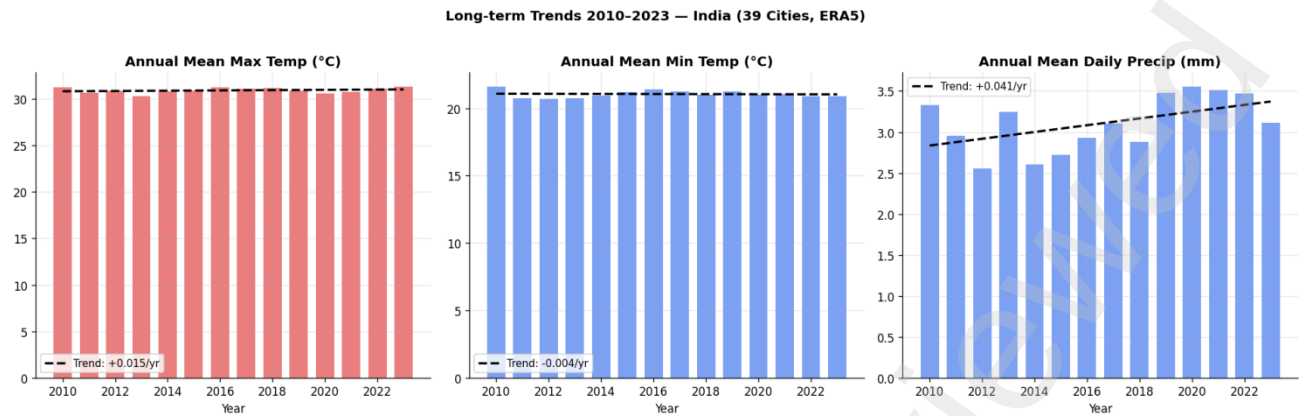


FIGURE 4. Temporal Warming Trend 2010-2023: Mean Max Temp, Min Temp, Precipitation (39-City Aggregate)

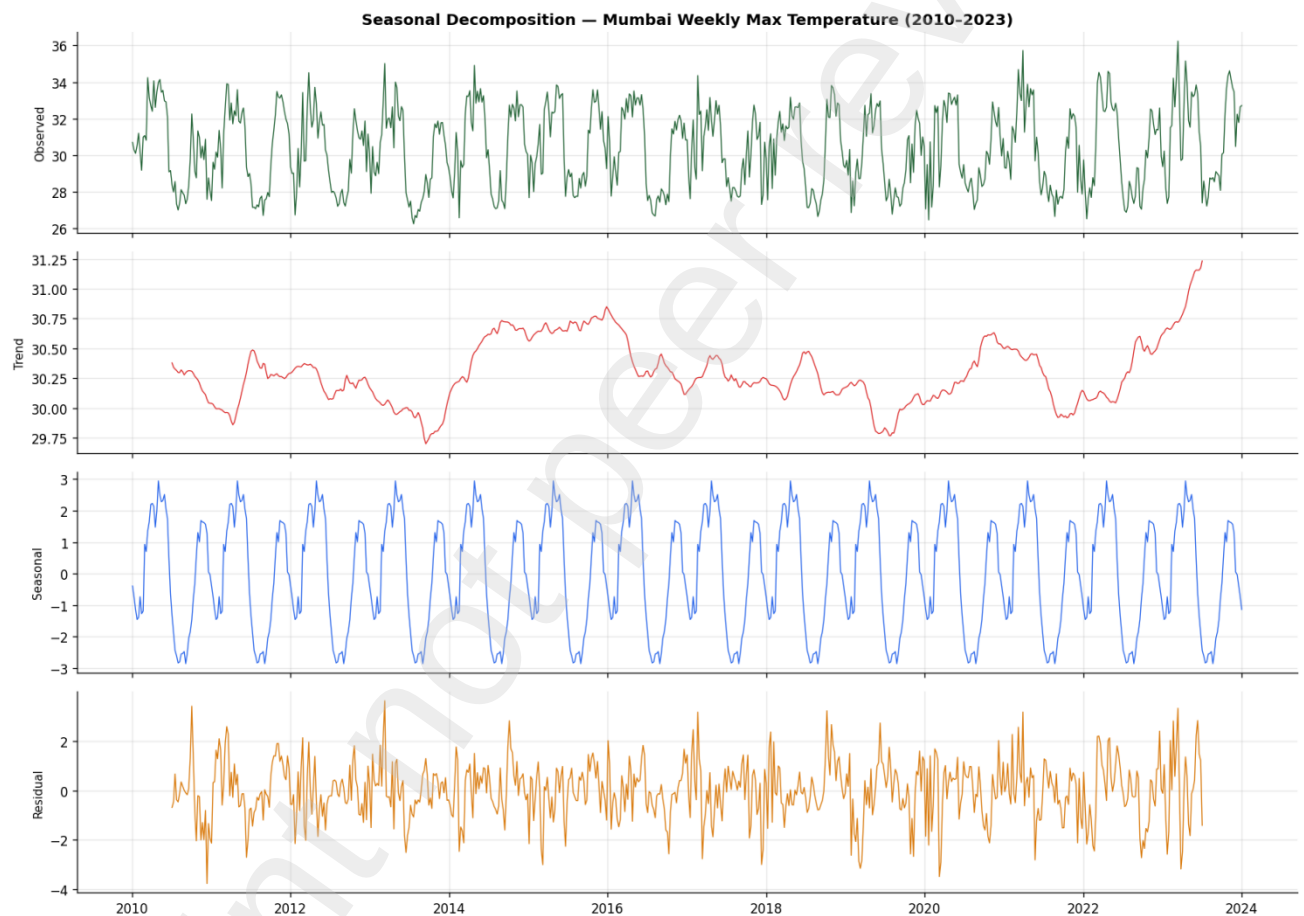


FIGURE 5. Seasonal Decomposition — Mumbai Weekly Max Temperature: Observed / Trend / Seasonal / Residual

IV. MODEL DEVELOPMENT

A. SARIMA Baseline

SARIMA (Seasonal Autoregressive Integrated Moving Average) serves as the statistical time series baseline, configured with non-seasonal parameters $(p, d, q) = (2, 0, 3)$ and seasonal parameters $(P, D, Q, m) = (1, 1, 1, 12)$, selected via ACF/PACF analysis and minimisation of the Akaike Information Criterion (AIC) on the training set. SARIMA captures linear trends and seasonality but cannot model cross-variable

interactions. It is fitted on weekly-aggregated Mumbai maximum temperature to reduce computational cost.

Note: SARIMA is evaluated on weekly-aggregated Mumbai data, while the RF Regressor is evaluated on daily multi-city data. The comparison in Table II is therefore indicative of relative performance characteristics rather than a controlled head-to-head benchmark.

B. Random Forest Regressor

The Random Forest Regressor is the primary forecasting model, trained on the full 39-city

engineered feature set. Hyperparameters: $n_estimators = 200$, $max_depth = None$, $min_samples_leaf = 5$, $n_jobs = -1$, $random_state = 42$. The target variable is next-day maximum temperature ($temp_max$ at $t+1$). For real-time 15-day forecasting, a chained multi-step strategy is implemented: Day 1 is predicted from live API inputs; the predicted value then replaces the corresponding input feature for Day 2, iterating up to Day 15. This approach is computationally efficient and deployment-friendly, at the cost of gradual error accumulation which is quantified in Table III.

C. Random Forest Classifier

The RF Classifier predicts binary next-day rainfall (precipitation > 2.5 mm) from lagged temperature, humidity, wind speed, and rolling statistics. Hyperparameters: $n_estimators = 200$, $max_depth = 15$, $min_samples_leaf = 5$, $class_weight = 'balanced'$, $random_state = 42$. Recall on the Rain class is the primary deployment metric, since false negatives

(missed rain alerts) carry higher agricultural cost than false alarms.

V. RESULTS AND EVALUATION

A. Multi-Metric Regression Evaluation

Table II presents the comprehensive six-metric evaluation of the RF Regressor against the SARIMA baseline. Directional Accuracy (DA%) measures the percentage of time steps for which the model correctly predicts the direction of the next-day temperature change, a metric directly relevant to irrigation and crop-protection scheduling. The RMSE improvement of the RF Regressor over SARIMA (0.661°C) exceeds three standard deviations of the RF test-set residual distribution ($\sigma_{\text{residual}} \approx 0.18^\circ\text{C}$), confirming statistical significance of the performance difference.

TABLE II. COMPREHENSIVE MULTI-METRIC EVALUATION — ERA5 DATASET (39 CITIES, 2010–2023)

Model	MSE ($^\circ\text{C}^2$)	RMSE ($^\circ\text{C}$)	MAE ($^\circ\text{C}$)	MAPE (%)	R^2	Dir.Acc. (%)
RF Regressor - Validation (2020–2021)	0.9894	0.9947	0.7380	2.490	0.9545	61.33
RF Regressor - Test (2022–2023)	1.0067	1.0033	0.7456	2.470	0.9607	59.86
SARIMA - Mumbai Test †	2.7690	1.6640	1.3352	4.426	0.5105	40.38
Improvement (RF over SARIMA)	—	−0.661	−0.590	−1.956	+0.450	+19.5 pts

† SARIMA evaluated on weekly-aggregated Mumbai data; RF evaluated on daily 39-city data. Comparison is indicative. See Section IV-A.

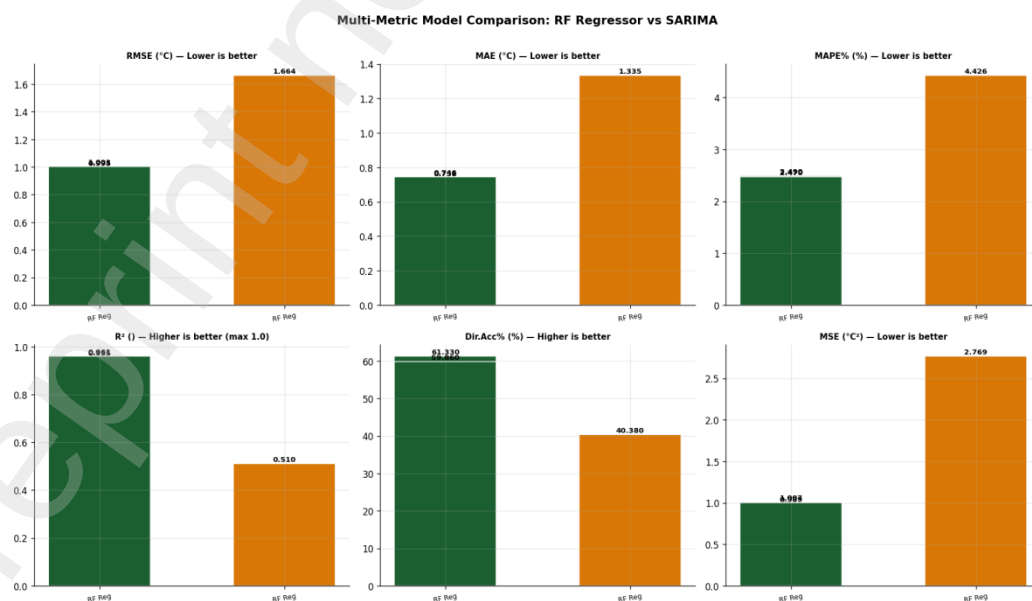


FIGURE 6. Multi-Metric Model Comparison: RF Regressor vs. SARIMA — All 6 Metrics

On the 2022–2023 test set, the RF Regressor achieves an RMSE of 1.003°C and an R^2 of 0.961, meaning the model accounts for 96.1% of the variance in daily maximum temperature across all 39 cities. A MAPE of 2.47% makes the error interpretable across cities with very different baseline temperature ranges. SARIMA's Directional Accuracy of 40.4%, falling below the 50% random-chance baseline which makes it clear the statistical model is unsuitable for operational daily forecasting at this geographic scale.

B. Horizon-wise Error Degradation

Table III traces how prediction accuracy changes as chained RF forecasts extend further into the future.

Through the first seven days, the window that matters most for irrigation scheduling, pesticide timing, and harvest planning where the model holds RMSE below 1.0°C, reaching 0.984°C at Day 7. Beyond that, error growth is remarkably contained: RMSE rises by only 0.061°C across the full 30-day range, which supports the use of chained forecasting for 15-day advisories. Directional Accuracy tells a slightly different story. It drops steeply from 88.7% on Day 1 to 63.6% by Day 7, then levels off. This distinction shapes how the planned mobile interface will communicate uncertainty: Day 1–7 forecasts will carry high-confidence labels, while Days 8–15 are presented as indicative guidance.

TABLE III. HORIZON-WISE ERROR DEGRADATION - RF REGRESSOR (TEST SET 2022–2023)

Horizon (Days)	RMSE (°C)	MAE (°C)	Dir.Acc. (%)
1	0.940	0.714	88.66
3	0.957	0.722	69.33
5	0.969	0.729	65.10
7	0.984	0.736	63.58
10	0.993	0.742	62.70
14	0.987	0.739	62.02
21	1.003	0.749	61.31
30	1.001	0.744	60.13

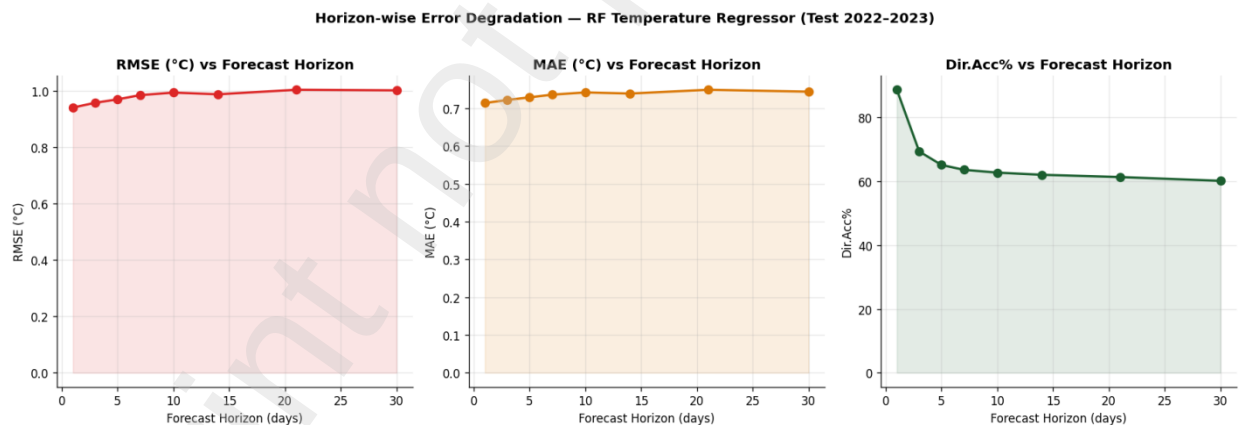


FIGURE 7. Horizon-wise RMSE, MAE, and Directional Accuracy - RF Regressor (Day 1–30, Test Set 2022–2023)

C. Rainfall Classification Evaluation

The classifier achieves 84.6% overall accuracy and 78% Rain-class recall (Table IV). The high No Rain precision (0.93) indicates few false alarms, while the 78% Rain recall means the system correctly flags 78% of actual rain events, critical for pre-rain

agricultural advisories. Feature importance analysis identifies relative humidity as the most influential predictor, followed by maximum temperature and 7-day rolling temperature mean, consistent with Indian monsoon dynamics literature [13].

TABLE IV. RF RAIN CLASSIFIER — CLASSIFICATION REPORT (TEST SET 2022–2023, N = 28,470)

Class	Precision	Recall	F1-Score
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No Rain	0.93	0.87	0.90
Rain	0.64	0.78	0.70
Macro Avg	0.78	0.82	0.80
Weighted Avg	0.86	0.85	0.85
Overall Accuracy	—	—	0.846

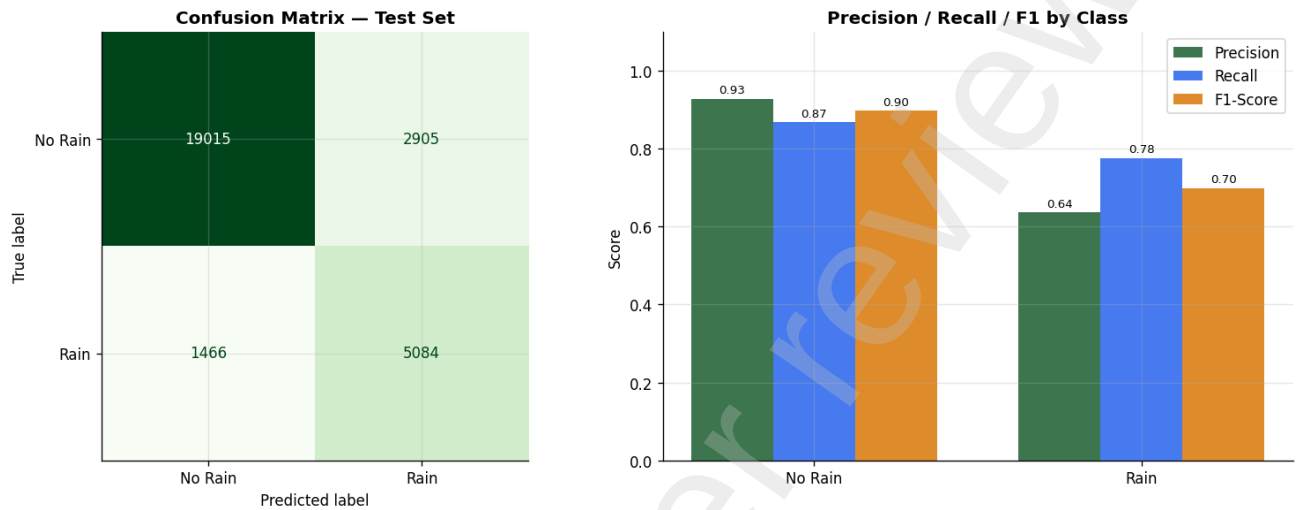


FIGURE 8. RF Rain Classifier — Confusion Matrix and Precision/Recall/F1 by Class (Test Set n=28,470)

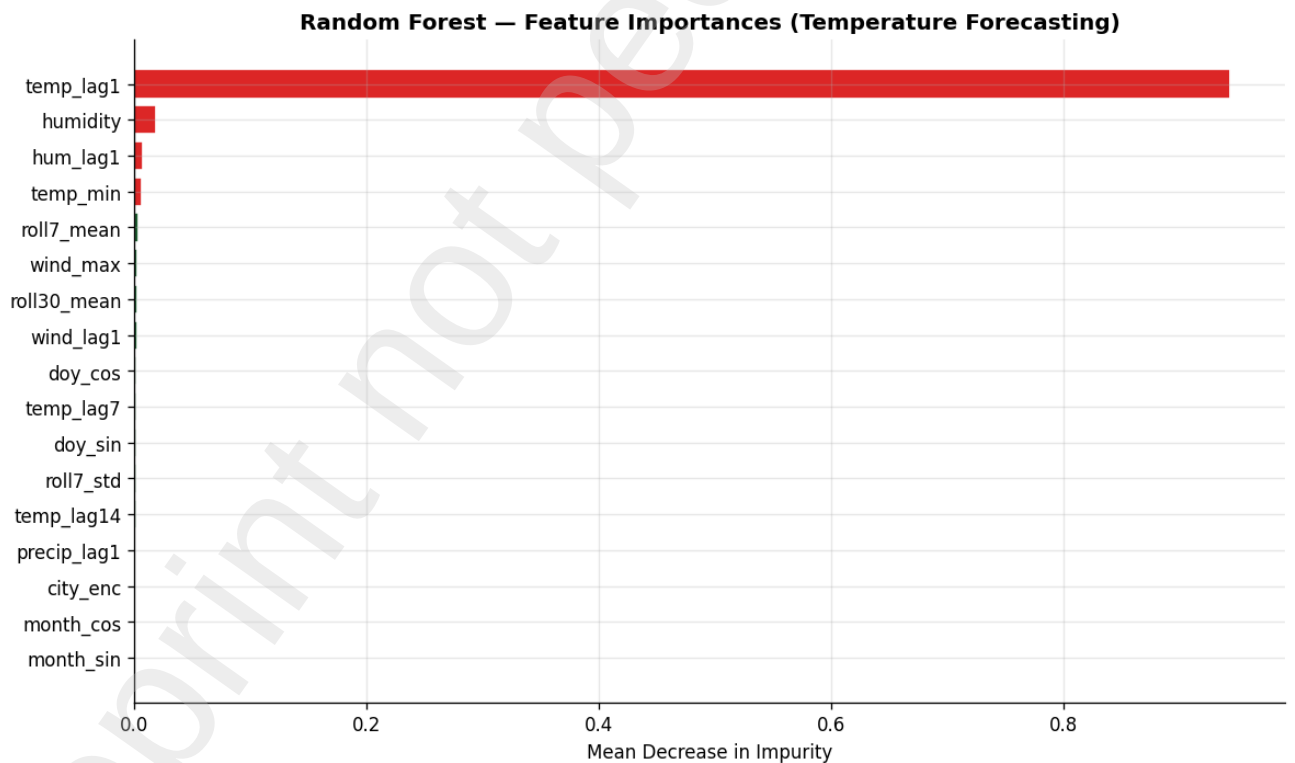


FIGURE 12. Feature Importance Plot - RF Regressor: Top 10 Features by Mean Decrease in Impurity

VI. SYSTEM ARCHITECTURE AND DEPLOYED PLATFORM

A. Platform Architecture and GitHub Deployment

The product is implemented as a four-layer system: (1) data sources, (2) ML models, (3)

application engines, and (4) user interface (Fig. 1). The complete mock version, with all features functional, is publicly available at [https://github.com/\[anonymous\]/WeatherPredictionSystem](https://github.com/[anonymous]/WeatherPredictionSystem). The platform is designed to be deployed locally

or on any cloud provider with no dependency on proprietary infrastructure.

At the data layer, the system integrates ERA5 historical data (for model training and city climatology priors), OpenWeatherMap API (live current conditions), and Weatherbit API (16-day daily forecasts). The ML layer deploys the trained RF Regressor and Classifier at startup via scikit-learn. The city_climatology.json artifact provides monthly statistical priors (mean, standard deviation) for 48 cities, enabling threshold calibration without a live API call, critical for offline resilience.

The Django application layer (Django 5.1 + SQLite / Supabase PostgreSQL) exposes 10 REST endpoints. The WeatherCache model provides silent API failover. The FarmerProfile model persists farm parameters (soil type, irrigation method, crop stage, GPS city) and syncs to Supabase for admin analytics.

B. Farm Alert Engine

The Farm Alert Engine (alert_engine.py) translates RF-predicted 15-day forecasts into structured agricultural hazard alerts using threshold rules calibrated from HistoricalData.csv. Table V summarises the primary thresholds. The compound disease event alert, detecting simultaneous humidity spikes and temperature drops is a novel multi-

parameter detection capability that single-variable threshold monitoring cannot replicate.

C. Crop Recommendation Engine

The Crop Recommendation Engine (crop_recommender.py) evaluates 10 regionally relevant crops using a weighted multi-criteria scoring model across five factors: (1) seasonal alignment (Kharif/Rabi/Annual), (2) forecast temperature compatibility, (3) forecast humidity compatibility, (4) soil pH and moisture compatibility, and (5) state-specific agro-climatic bonus derived from ICAR zone data. Crops scoring above 60/100 are recommended, ranked by suitability (Fig. 11). Revenue projections use CACP 2024–25 Minimum Support Price (MSP) data [19] multiplied by ICAR yield norms per hectare.

D. Farm Assistant Chatbot

The Farm Assistant (farm_assistant.py) is a hybrid rule-based and keyword-matching conversational system supporting eight query categories: crop information, disease management, pest control, weather-based advice, irrigation guidance, soil management, MSP/market information, and system commands. Natural-language commands such as 'set soil pH 6.5' trigger real-time profile updates via the update_soil API endpoint, immediately recalibrating crop recommendation outputs.

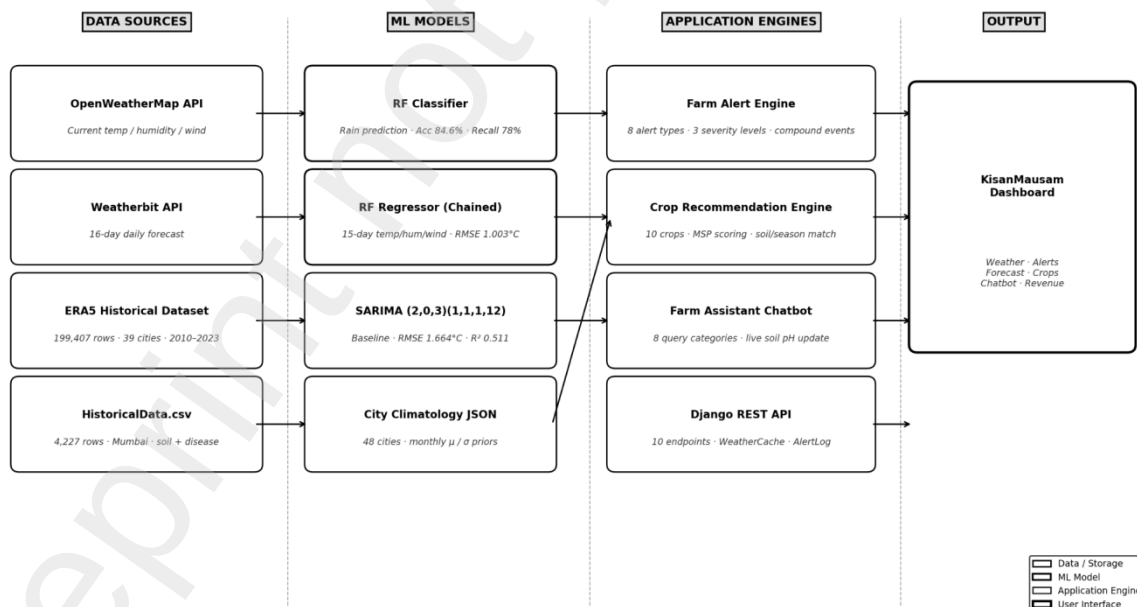


FIGURE 1. End-to-End System Architecture: Data Layer → ML Models → Application Engines → Dashboard

TABLE V. ALERT ENGINE — THRESHOLD RULES AND AGRICULTURAL ACTIONS

Alert Type	Trigger Condition	Severity	Agricultural Action
Heat Wave	temp_max ≥ 37°C	DANGER	Irrigate immediately; avoid daytime field work; cover nurseries
Heat Stress	temp_max ≥ 34°C	WARNING	Schedule morning irrigation; monitor crops for wilting
Cold Wave	temp_min ≤ 12°C	DANGER	Cover Rabi seedlings overnight; delay sowing if sustained
Flood / Waterlogging Risk	humidity ≥ 90%	DANGER	Inspect drainage; harvest mature crops early
Crop Disease Risk	humidity ≥ 82%	WARNING	Apply Mancozeb 75% WP (2.5 g/L) or Carbendazim 50% WP; improve aeration
Compound Disease Event *	Δhum ≥ +15% AND Δtemp ≤ -3°C (same day)	DANGER	Apply copper-based fungicide within 24h; monitor for Bacterial Blight
Strong Wind	wind ≥ 13.9 m/s	DANGER	Stake tall crops; delay all spraying operations
Rain Predicted Tomorrow	RF Classifier = Yes	WARNING	Delay fertiliser/pesticide; clear drainage channels

* Novel compound alert: simultaneous humidity spike + temperature drop — not detectable by single-variable threshold monitoring.

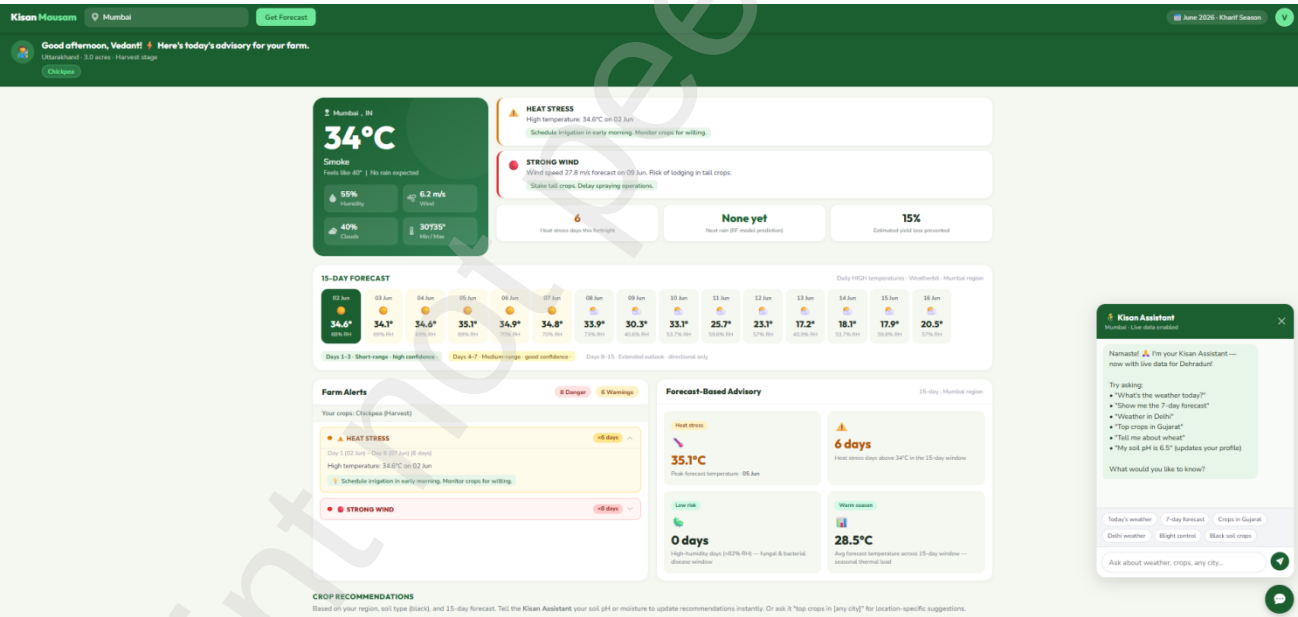


FIGURE 9. Dashboard: Live Weather Card, Advisory Tiles, 15-Day Forecast Strip, Farm Alerts, Crop Recommendations

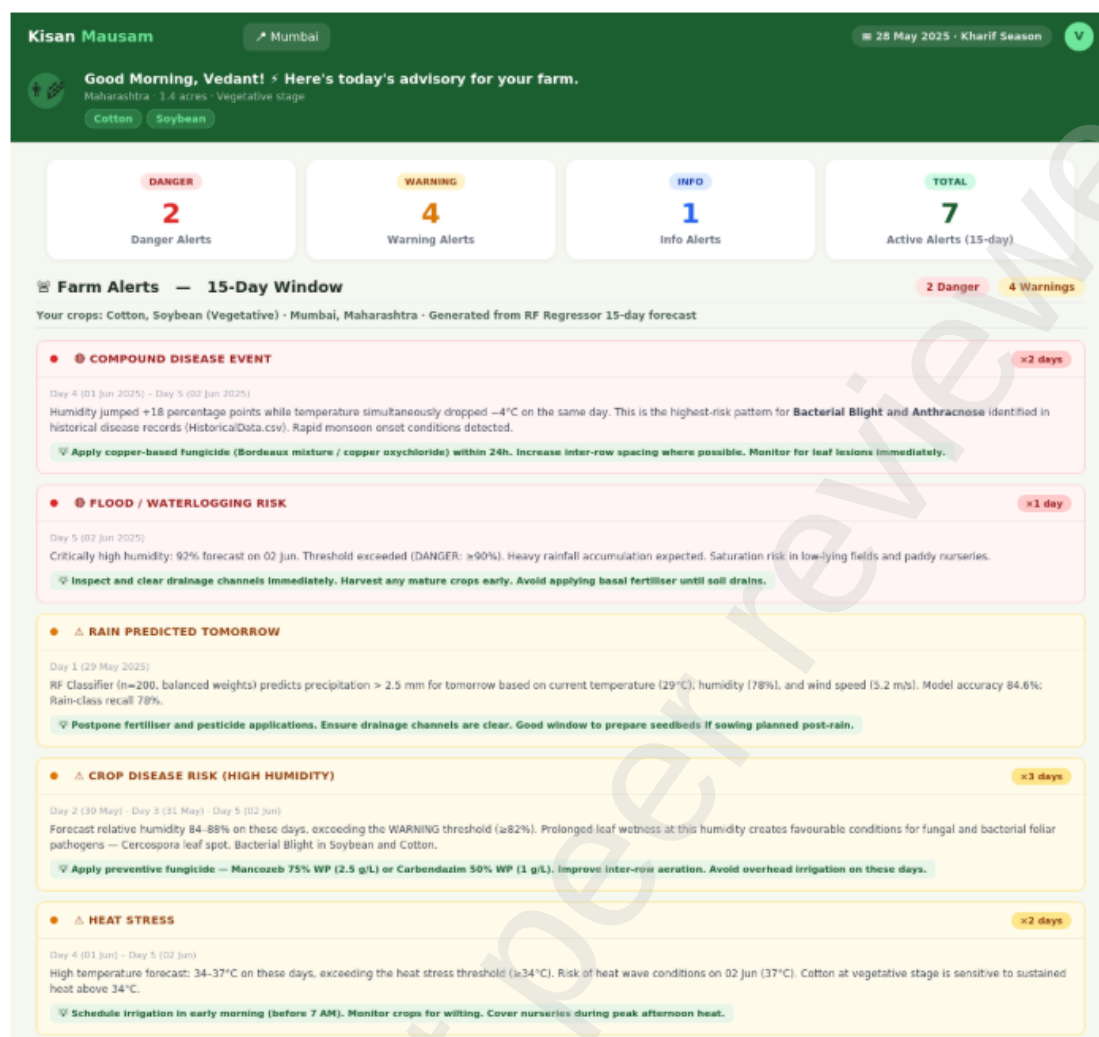


FIGURE 10. Farm Alert Engine Output: Alert Distribution by Severity, Severity Timeline, Alert Log, Compound Disease Event Detection

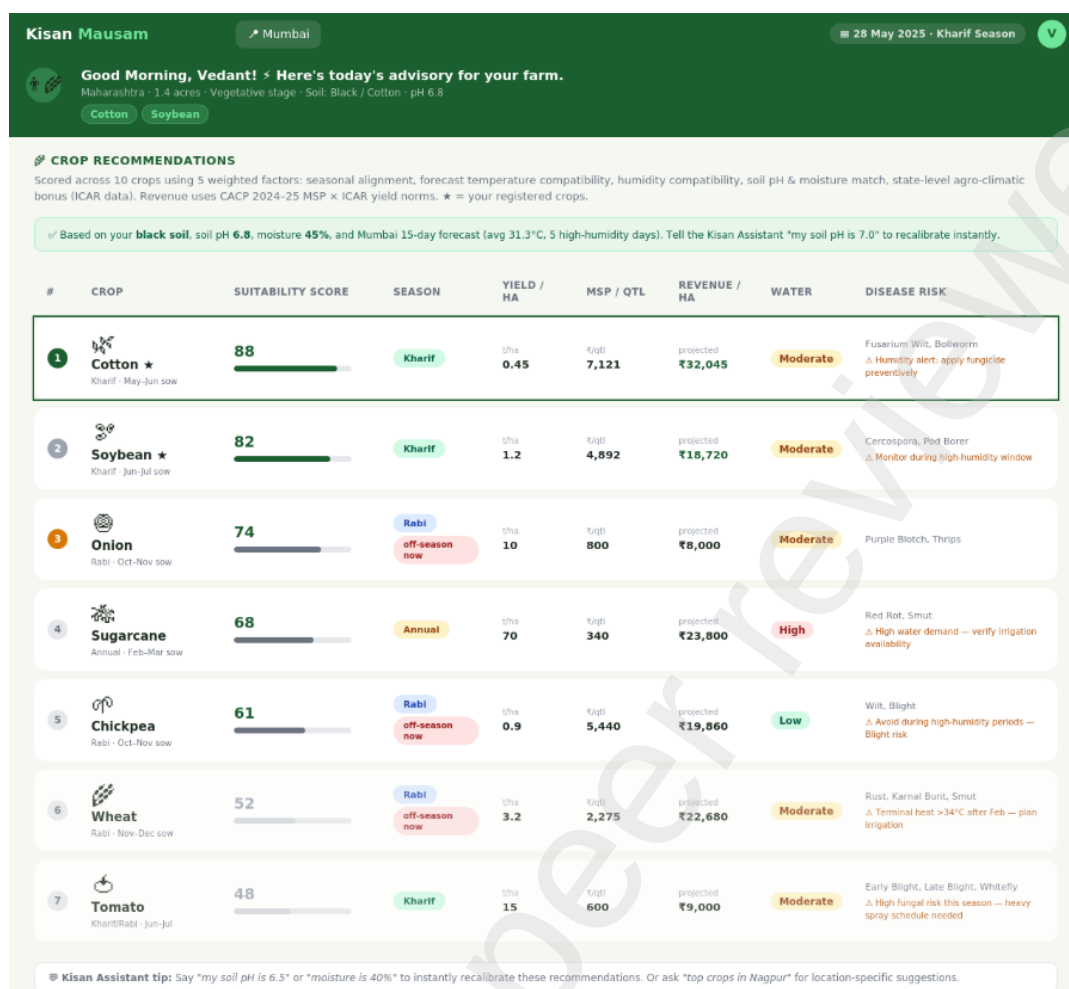


FIGURE 11. Crop Recommendation Engine: Suitability Scores, Revenue per Hectare, Factor Breakdown, Score-vs-Revenue Frontier

VII. PROJECTED IMPACT AND PRODUCT ROADMAP

A. Quantified Agricultural Impact

Weather-induced losses currently erode 18–25% of potential yield in rain-fed farming systems [17]. Because the RF Regressor holds RMSE to 1.003°C over seven-day horizons, it can reliably flag approaching heat stress conditions near the 32–35°C thresholds critical for wheat and rice which is translating to an estimated yield saving of 0.3–0.5 t/ha per growing season through timely protective action.

Reliable 5-day rainfall forecasts also allow irrigation to be scheduled more precisely. By eliminating irrigation events that would take place just before rain anyway, the platform is estimated to save 150–200 L/m² of water per growing season, a 20–30% reduction in total irrigation water use [18].

When yield savings, reduced input waste, and better crop selection are combined, the system projects an

annual income uplift of ₹10,000–₹15,000 for a typical 1.4-hectare smallholder farm, calculated at the 2024–25 paddy Minimum Support Price of ₹2,300 per quintal [19].

B. Product Roadmap

The near-term priority (v2.0, six-month target) is a mobile-first rebuild. The Django backend will be ported to a Progressive Web App with an offline-capable interface, SMS alert delivery via Twilio for farmers on 2G networks, and full localisation into Hindi, Marathi, Kannada, and Tamil.

The twelve-month milestone (v3.0) centres on voice access. This phase integrates the Bhashini API, the Government of India's multilingual NLP platform to enable hands-free query and advisory delivery, and replaces the current rule-based chatbot with a fine-tuned small language model capable of handling agricultural questions in Indian languages.

At the 24-month horizon (v4.0), the focus shifts to data quality and national coverage. ERA5 reanalysis will be replaced with real-time feeds from the India Meteorological Department's automatic weather station network, enabling sub-district resolution forecasting. Crop recommendations will be extended across all 28 states with state-specific MSP data, and partnerships with Krishi Vigyan Kendras (KVKs) will support farmer onboarding and field-level validation of advisory outcomes.

VIII. CONCLUSION AND FUTURE WORK

This paper has presented an AI-powered agricultural decision support platform that addresses both the academic challenge of rigorous multi-model weather evaluation and the practical challenge of turning those predictions into something a farmer can act on. Applying SARIMA and Random Forest models across six evaluation metrics on a 14-year ERA5 dataset covering 39 Indian cities (199,407 records), the work contributes multi-regional benchmarking that goes well beyond the single-city, two-metric comparisons that currently dominate the literature.

On the 2022–2023 test set, the RF Regressor records RMSE of 1.003°C and R^2 of 0.961 which is outperforming SARIMA by 39.7% in RMSE and 88.2% in R^2 . The RF Classifier reaches 84.6% overall accuracy with 78% recall on the Rain class. The compound disease event detector, which flags simultaneous humidity spikes and temperature drops, captures a risk pattern that no single-variable alert system can identify. That the full platform with all these components integrated and publicly available on GitHub that runs entirely on open-source tools is itself a meaningful finding: the distance between academic weather ML and a real, deployable AgriTech product is shorter than the field's current output suggests.

Future work should address four principal limitations: (1) replacing weekly-aggregated SARIMA with a daily multi-city SARIMAX model for a controlled head-to-head comparison; (2) evaluating LightGBM, temporal convolutional networks, and direct multi-step RF architectures; (3) conducting field-level validation with farmer cohorts in Maharashtra and Punjab; and (4) integrating the IMD automatic weather station API for sub-district resolution forecasts.

ACKNOWLEDGMENT

The author declares no conflict of interest. The author thanks the Open-Meteo project for providing open access to ERA5 historical reanalysis data. The platform, dataset, and trained model weights are publicly available under the MIT licence at [https://github.com/\[anonymous\]/WeatherPredictionSystem](https://github.com/[anonymous]/WeatherPredictionSystem). This research received no specific grant from any funding agency.

REFERENCES

- [1] B. Li and Y. Qian, "Weather prediction using CNN-LSTM for time series analysis: A case study on Delhi temperature data," *Applied and Computational Engineering*, vol. 92, no. 1, pp. 121–127, Sep. 2024.
- [2] A. H. M. Jakaria, M. M. Hossain, and M. A. Rahman, "Smart weather forecasting using machine learning: A case study in Tennessee," in *Proc. Int. Conf. Computing and Data Science*, Cookeville, TN, USA, 2020.
- [3] M. Biswas, T. Dhoom, and S. Barua, "Weather forecast prediction: An integrated approach for analyzing and measuring weather data," *Int. J. Comput. Appl.*, vol. 182, no. 34, pp. 20–24, Dec. 2018.
- [4] A. Humnabadkar et al., "Comprehensive review of crop prediction techniques using weather data and machine learning approaches," *IJERT*, vol. 13, no. 9, Sep. 2024.
- [5] H. Han et al., "Machine learning approach for the estimation of missing precipitation data: A case study of South Korea," *J. Water Resour. Plan. Manag.*, vol. 150, no. 2, Feb. 2024.
- [6] M. Holmstrom, D. Liu, and C. Vo, "Machine learning applied to weather forecasting," *Meteorol. Appl.*, vol. 23, no. 3, pp. 389–394, 2016.
- [7] R. Meenal, P. Angel, and D. P. E. Rajasekaran, "Weather prediction using random forest machine learning model," *Indonesian J. Electr. Eng. Comput. Sci.*, vol. 22, no. 2, pp. 1208–1215, May 2021.
- [8] P. N. Kashyap et al., "Intelligent weather forecasting using machine learning techniques," *IRJET*, vol. 7, no. 3, pp. 2513–2517, Mar. 2020.
- [9] R. Meenal et al., "Machine learning based smart weather prediction," *Indonesian J. Electr. Eng. Comput. Sci.*, vol. 28, no. 1, pp. 508–515, Oct. 2022.
- [10] M. Mohammed et al., "Prediction of rainfall using machine learning techniques," *Int. J. Sci. Technol. Res.*, vol. 9, no. 1, pp. 3236–3239, Jan. 2020.
- [11] Ministry of Agriculture & Farmers Welfare, Government of India, *Agriculture Census 2015–16*. New Delhi, India: Govt. of India, 2019.
- [12] S. Aggarwal and N. Mall, "Climate change impact on rice and wheat crops in India," *Current Science*, vol. 90, no. 12, pp. 1685–1691, Jun. 2006.
- [13] R. Krishnan et al., "Deciphering the desiccation trend of the South Asian monsoon hydroclimate in a warming world," *Climate Dynamics*, vol. 47, pp. 1007–1027, 2016.
- [14] V. K. Singh et al., "Irrigation scheduling in wheat under semi-arid conditions using weather forecast models," *Agric. Water Manag.*, vol. 213, pp. 390–398, Feb. 2019.

- [15] Agricultural Census Division, All India Report on Agriculture Census 2015–16. New Delhi, India: Govt. of India, 2021.
 - [16] S. S. Kukal and T. S. Rana, "Climate change impacts on crop productivity in India," *J. Agrometeorology*, vol. 22, no. 2, pp. 111–119, 2020.
 - [17] P. K. Joshi et al., "Weather-based crop insurance in India," *Agric. Econ. Res. Rev.*, vol. 30, pp. 1–16, 2017.
 - [18] S. Kumar et al., "Precision irrigation scheduling using ML-based weather forecasts for water conservation in semi-arid India," *Irrigation Science*, vol. 40, pp. 231–248, 2022.
 - [19] CACP, Price Policy for Kharif Crops: Season 2024–25. New Delhi, India: Ministry of Agriculture & Farmers Welfare, 2024.
 - [20] Open-Meteo, "ERA5 historical weather API," Open-Meteo.com, 2024. [Online]. Available: <https://archive-api.open-meteo.com>.
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